

boilers	3 rows	operators	6 rows	water_tests	15 rows
one row = one boiler · the asset register		one row = one person on the roster		one row = one weekly water sample	
boiler_id PK	internal id, 1–3	operator_id PK	1 ... 6	test_id PK	1 ... 15
code	plant tag 'B-2101'	emp_no	'EM-4471'	boiler_id FK	→ boilers.boiler_id
name	'North watertube'	full_name	'Somchai Srisuk'	test_date	'YYYY-MM-DD'
maker_model	'Babcock FM-120'	shift	'A', 'B' or 'C'	silica_ppm	ppm (plant limit 10)
rating_tph	design output, t/h	cert_class	'Class 1' / 'Class 2'	conductivity_us	µS/cm
design_barg	design pressure, barg	Values in 'quotes' are text — everything else is a number.		phosphate_ppm	residual, ppm
commissioned	year in service			ph	pH of the sample
fuel	'Natural gas'				

readings	540 rows	daily_logs	90 rows	clause order	always this order
one row = one instrument reading, every 4 h		one row = one boiler on one day		WITH	a named sub-query optional
reading_id PK	1 ... 540	log_id PK	1 ... 90	SELECT	which columns you want
boiler_id FK	→ boilers.boiler_id	boiler_id FK	→ boilers.boiler_id	FROM	which table
ts	'YYYY-MM-DD HH:MM'	operator_id FK	→ operators.operator_id	JOIN...ON	how another table matches
steam_tph	steam flow, t/h	log_date	'YYYY-MM-DD'	WHERE	which rows to keep
stack_temp_c	flue gas, °C	steam_t	steam that day, t	GROUP BY	what to summarise by
o2_pct	flue gas O ₂ , %	fuel_gj	gas that day, GJ	HAVING	which groups to keep
				ORDER BY	how to sort DESC = big first
				LIMIT	how many rows to show
				Only SELECT and FROM are compulsory.	

boiler_id is not the tag. 1 = B-2101 North · 2 = B-2103 South · 3 = B-2102 Centre — the big tables store only the id, so never name a machine from its number. JOIN boilers b ON b.boiler_id = r.boiler_id and read the tag.

From the question to the query

"How many ...?"	→ COUNT(*) · COUNT(DISTINCT x)	"The worst / the top five"	→ ORDER BY x DESC + LIMIT 5
"For each boiler / shift / day"	→ GROUP BY that column	"Show the tag, not the id"	→ JOIN the small table
"Only where ..."	→ WHERE rows · HAVING groups	"How often is it above X?"	→ SUM(CASE WHEN x>X THEN 1 ELSE 0 END)
"Per tonne / per hour / per day"	→ divide one SUM by another	"Is it getting worse?"	→ GROUP BY strftime('%W', ts)

Traps that catch everybody

- Text needs **single** quotes:
code = 'B-2103', never "B-2103".
Numbers take none: boiler_id = 2.
- WHERE filters rows, HAVING filters groups —
WHERE AVG(x) > 150 is an error.
- Write 100.0, not 100. Whole numbers
divide into whole numbers, so 3/4 is 0 and
percentages come out zero.
- A rate for a whole month is
SUM(fuel_gj)/SUM(steam_t), **not**
AVG(fuel_gj/steam_t) — the second lets
a quiet day count as much as a busy one.
- ts holds a date *and* a time, so match the
start: ts LIKE '2026-04-01%'.
Anything in SELECT beside an aggregate
must also appear in GROUP BY.
- Put COUNT(*) next to every average — it
says how many rows went into it.

The shape of a real query

```
-- gas per tonne of steam, worst boiler first
SELECT  b.code,
        ROUND(SUM(d.fuel_gj) / SUM(d.steam_t), 3) AS gj_per_tonne,
        COUNT(*) AS n_days
FROM    daily_logs d
JOIN    boilers b ON b.boiler_id = d.boiler_id
WHERE   d.log_date LIKE '2026-04%'
GROUP BY b.code
HAVING  COUNT(*) > 25
ORDER BY gj_per_tonne DESC

Pieces worth remembering: AVG MIN MAX SUM COUNT · ROUND(x, 1) · substr(ts,
1, 10) the day · strftime('%W', ts) the week · x IN ('B', 'C') · (SELECT
AVG(x) FROM t) used as one value
```